

TOPCLIFFE, United Kingdom

WMO: 032650

Lat: 54.2047N

Lon: 1.3900W

Elev: 92

StdP: 14.65

Time Zone: 0.00 (EUW)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.2	25.9	20.3	15.3	26.6	22.8	17.2	29.1	33.2	47.9	29.8	45.8	4.2	130	0.643

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	76.7	65.4	73.3	63.3	70.4	61.7	66.9	73.6	64.8	70.8	63.0	68.5	10.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.3	90.6	70.1	62.4	84.6	67.6	60.8	79.7	65.6	31.5	73.8	29.9	70.8	28.6	68.7	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.1	23.4	20.8	16.3	83.3	5.1	3.5	12.6	85.8	9.6	87.9	6.8	89.8	3.1	92.3		
(5)			16.1	71.3	5.1	3.1	12.4	73.6	9.4	75.4	6.6	77.2	2.9	79.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.8	39.0	39.7	42.4	46.4	52.1	56.9	60.3	60.0	56.0	50.2	43.3	39.2		
	DBStd	9.45	5.82	5.83	5.20	5.08	5.10	4.57	4.21	4.25	4.37	5.32	5.91	7.09		
	HDD50	1658	341	290	241	130	37	3	0	0	4	62	211	339		
	HDD65	5923	806	708	701	557	401	248	156	165	271	459	651	800		
	CDD50	1235	1	2	5	23	100	208	320	310	185	68	10	3		
	CDD65	28	0	0	0	0	0	4	12	10	2	0	0	0		
	CDH74	232	0	0	0	1	7	39	109	62	13	1	0	0		
	CDH80	25	0	0	0	0	0	4	13	8	0	0	0	0		
	WSAvg	9.7	10.6	10.5	10.4	9.7	10.0	9.0	8.5	8.9	8.8	9.7	9.9	10.0		
	PrecAvg	27.0	2.4	1.9	1.6	2.0	1.9	2.4	2.2	2.7	2.2	2.5	2.7	2.6		
(15) Precipitation	PrecMax	37.8	5.0	3.5	3.5	5.5	3.8	6.4	5.0	6.1	4.5	5.8	5.4	5.1		
	PrecMin	20.4	0.3	0.4	0.4	0.2	0.4	0.3	0.7	0.6	0.6	0.8	0.9	0.8		
	PrecStd	4.3	1.1	0.9	0.9	1.4	0.8	1.5	1.0	1.3	1.2	1.1	1.1	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.0	56.6	61.0	68.7	74.3	79.0	81.2	80.9	75.8	66.5	59.8	56.2		
		MCWB	51.8	52.3	52.3	57.0	64.2	67.8	65.2	68.2	64.8	60.9	56.4	53.0		
	2%	DB	52.6	53.9	57.2	62.7	69.8	73.3	77.1	74.7	70.7	63.0	56.9	54.0		
		MCWB	49.4	50.2	50.5	54.4	60.7	62.7	65.0	64.8	62.1	58.4	53.6	51.1		
	5%	DB	50.4	51.4	54.3	59.3	65.7	69.9	73.5	71.1	67.1	60.6	54.9	52.1		
		MCWB	47.1	47.8	49.1	52.0	57.4	60.9	63.4	62.6	59.9	56.7	51.7	49.5		
	10%	DB	48.4	49.3	52.0	56.3	62.2	66.8	69.9	68.6	64.5	58.7	52.6	50.0		
		MCWB	45.5	45.9	47.1	50.3	55.0	59.1	61.5	60.9	58.2	55.1	49.8	47.4		
	0.4%	WB	52.2	52.7	53.9	58.9	66.0	70.2	69.0	69.5	67.1	62.4	57.2	53.7		
		MCDB	54.5	55.7	58.5	64.8	70.1	75.1	77.6	78.4	73.8	65.8	59.5	55.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	49.8	50.7	51.7	55.5	61.8	64.7	66.4	66.2	63.5	59.6	54.3	51.8		
		MCDB	52.1	53.5	55.9	61.2	68.5	70.2	73.8	72.4	68.5	61.8	56.3	53.6		
	5%	WB	47.8	48.4	49.7	53.1	58.8	61.9	64.4	64.3	61.2	57.6	52.2	49.9		
		MCDB	50.1	51.0	53.7	58.1	64.6	67.7	71.0	69.4	65.2	60.0	54.3	51.9		
	10%	WB	45.7	46.3	47.7	50.9	56.0	60.0	62.5	62.5	59.4	55.7	50.2	47.6		
		MCDB	48.0	49.0	51.6	55.6	60.9	65.4	68.8	66.9	63.3	58.1	52.3	49.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	12.2	14.7	16.4	16.8	16.4	16.7	16.4	15.9	13.3	11.8	11.1		
		MCDBR	12.3	13.8	18.7	21.6	23.0	22.9	23.2	19.8	20.3	14.8	12.9	12.2		
		MCWBR	11.0	11.3	13.1	14.0	15.0	13.8	12.3	11.2	12.8	11.0	10.7	10.8		
	5% WB	MCDBR	12.4	13.2	17.3	19.9	21.6	20.2	20.3	17.3	17.8	14.1	12.0	11.8		
		MCWBR	11.3	11.3	13.0	13.9	15.6	14.0	12.2	10.8	12.8	11.5	10.6	10.8		
(40) Clear-Sky Solar Irradiance	taub		0.321	0.337	0.368	0.396	0.395	0.398	0.399	0.386	0.366	0.348	0.322	0.307		
	taud		2.316	2.366	2.287	2.230	2.277	2.293	2.298	2.349	2.411	2.431	2.378	2.293		
	Ebn at Noon		198	238	254	263	270	270	267	263	254	231	198	176		
	Edh at Noon		18	24	32	39	39	39	38	34	29	23	17	16		
(44) All-Sky Solar Radiation	RadAvg		196	399	743	1171	1493	1523	1471	1215	882	492	242	151		
	RadStd		23	46	95	130	87	134	134	79	80	42	24	15		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page